STATISTICS IS THE GRAMMAR OF SCIENCE

PROBABILITY AND STATISTICS

LECTURE – 29

LOGISTIC REGRESSION

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QUALITATIVE RESPONSE REGRESSION MODELS

Economics is sometimes described as the "theory of choice" Many choices we make in life are 'either' – 'or' in nature. A few examples include.

1. A consumer who choose between Coke and Pepsi.
2. A bank official must choose to Accept a loan application or not.
3. A High school graduate must decide whether to attend college or not.
4. A worker decides either to drive to work or to get there using different means of transportation.
5. A Household decides either to purchase a house or to rent.
6. A firm decides either to advertise its product in a local newspaper or it decides not to.
7. A Married woman who must decide whether to enter the labor market or not.
8. What determines a teenager takes up smoking, the teenager does or does not take up smoking.
9. What determines whether a job applicant is successful? the applicant does or does not get a job.
10. What determines whether a country receives foreign aid, a country does or does not receive foreign aid.

⊛→ As an economists we are interested in explaining why particular choices are made and what factors enter into the decision process. we also want to know

⊛ In all these examples the outcome of interest is binary. In each case decision maker chooses [illegible] two mutually exclusive outcomes.

how much each factor affects the outcome. Such questions lead us to the problem of constructing a statistical model of binary, either-or, choices.

To analyze and predict such outcomes using an econometric model, we represent the choice using an indicator variable, the value one if one alternative is chosen and the value zero if the other alternative is chosen. Because we are attempting to explain choice between two alternatives, the indicator variable will be the dependent variable rather than an independent variable in a regression model.

We will illustrate binary choice models using an important problem from transportation economics. How can we explain an individuals choice between driving (Private transport) and taking the bus (Public transportation) when commuting to work. For simplicity assume that there are only two alternatives? We represent an individual's choice using an indicator variable.

$$y = \begin{cases} 1, & \text{individual drives to work.} \\ 0, & \text{individual takes bus to work.} \end{cases}$$

If we collect a random sample of workers who commute to work, then the outcome y will be unknown to us until the sample is drawn. Thus y is random variable.

If the probability that an individual drives to work is p, then $P(y=1) = p$. It follows that the probability that a person uses public transportation is $P(y=0) = 1-p$

The probability function for such a binary random variable is

$$f(y) = p^y \cdot (1-p)^{1-y}, \quad y = 0, 1$$

where 'p' is the probability that y takes the value one. We can see that discrete random variable y follows a Bernoulli distribution. Almost all the models having Binary dependent variables follow Bernoulli distribution. So $E(y) = p$, and $var(y) = p(1-p)$.

What factors might affect the probability that an individual chooses one transportation mode over the other? One factor will certainly be how long it takes to get to work one way or the other.

Define the explanatory variable.

$x = (\text{Commuting time by bus} - \text{commuting time by car})$

There are other factors that affect the decision, but let us focus on this single explanatory variable. A priori we expect that as 'x' increases, and commuting time by bus increases relative to commuting time by car, an individual would be more inclined to drive. i.e we expect a positive relationship between x and p, the probability that an individual will drive to work.

When the dependent variable is Binary, things are more difficult: what it mean to fit a line to a dependent variable that can take only two values 0 and 1. The answer to this question is to interpret the regression function as a predicted probability.

We start our study of qualitative response models by first considering Binary response regression Models there are four approaches to develop a probability model for binary response variables.

1- The Linear Probability Model.

2- The Logit Model

3- The Probit Model

4- The Tobit Model.

The Binary dependent variable is an example of dependent variable with limited range in other words, it is limited dependent variable. There are other Models having limited dependent variables that take on multiple discrete values, These models are called Polychotomous (Multiple-category) response variables Models. For example in general elections in Pakistan A voter may vote to PMLN, PTI or PPP (three parties) so here the response variable is trichotomous.

THE LOGIT REGRESSION MODEL

consider the regression model

$$Y_i = \beta_0 + \beta_1 X_i + u_i \longrightarrow (1)$$

where X = Family income

and $Y = \begin{cases} 1 & \text{if the family owns a house} \\ 0 & \text{if it does not own a house.} \end{cases}$

Then the population logit model of binary dependent variable Y with regressor X_i is

$$Pr(Y_i = 1 | X_i) = E(Y_i = 1 | X_i) = F(\beta_0 + \beta_1 X)$$

$$= \frac{1}{1 + e^{-(+\beta_0 + \beta_1 X_i)}}$$

$$\text{So } P_i = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_i)}} \longrightarrow (2)$$

Let $Z_i = \beta_0 + \beta_1 X_i$

So (2) becomes

$$P_i = \frac{1}{1 + e^{-Z_i}}$$

$$P = \frac{1}{1 + e^{-Z}} = \frac{e^Z}{1 + e^Z} \longrightarrow (3)$$

Equation (3) is known as logistic (cumulative) dist. Fn.

If 'P' is the probability of owning a house then '1-P' is the probability of not owning a house and is given by

$$1 - P = 1 - \frac{e^Z}{1 + e^Z} = \frac{1}{1 + e^Z} \longrightarrow (4)$$

Now $\frac{P}{1-P}$ is simply the odds ratio in favour of owning a house ie the ratio of the Prob. that family will own a house to the Prob it will not own a house. and it is

(7)

Given by

(Note odds = $\frac{P(y=1)}{P(y=0)}$
odds > 1 if y=1 is more likely, odds < 1 if y=0 is more likely)

$$\frac{P}{1-P} = \frac{e^z/1+e^z}{1/1+e^z} = e^z \longrightarrow (5)$$

Thus if P = 0.8

⇒ odds are 4 to 1 in favour of the family owning a house.

Now taking ln of RS of (5) we get.

$$\ln\left(\frac{P}{1-P}\right) = \ln(e^z) = z$$

$$L = \beta_0 + \beta_1 X \longrightarrow (6)$$

'L' is the log of odds ratio, it is not only linear in X but also linear in Parameters.

'L' is called the logit and hence the name logit model for models like Eqn (6).

PROPERTIES OF THE LOGIT MODEL

1- As P goes from 0 to 1, the logit L goes from $-\infty$ to $+\infty$.
⇒ Logits are unbounded.

2- Although 'L' is linear in X, the probabilities themselves are not.

3- LPM assumes that P_i is linearly related to X_i, the logit model assumes that the log of the odds ratio is linearly related to X_i.

4- one can add as many regressors as may be regd by the theory In that case Logit Model becomes

$$Pr(Y=1|X_1, X_2, \ldots, X_k) = F(\beta_0 + \beta_1 X_1 + \ldots + \beta_k X_k) = \frac{1}{1+e^{-(\beta_0+\beta_1 X_1+\ldots+\beta_k X_k)}}$$

5- If 'L' is +ve ⇒ when value of regressors increases, the odds that the regressand equals '1' increases. If L is -ve the odds that regressand equals 1 decreases as X's increases.

ESTIMATION OF LOGIT COEFFICIENTS

The Logit coefficients can be estimated using MLE Method. MLE chooses estimates of β_0 and β_1 that maximize the Probability of observing the sample. Software uses MLE for estimating Logit Models. The MLE estimator is consistent and Normally distributed in large samples, so that t-statistics and confidence intervals for the coefficients can be constructed in the usual way.

INTERPRETITION OF LOGIT COEFFICIENTS

The estimated coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$ are difficult to interpret because they affect the probabilities of dependent variable via z-value. The Logit coefficients are best interpreted by computing predicting probabilities and difference in Predicted Probabilities.

For Predicted Probabilities we use Logistic Response Function, $P(y=1) = \phi(\hat{\beta}_0 + \hat{\beta}_1 X)$

$$P(y=1) = \frac{1}{1+e^{-(\hat{\beta}_0 + \hat{\beta}_1 X)}}$$

For the given $X = X_0$ there is some Predicted Probab-

Also $P(y=0) = 1 - P(y=1)$

Positive values are predictive of class '1'.

Negative values are predictive of class '0'.

By comparing a threshold value, like 0.5, we can predict choice using the rule

$$\hat{y} = \begin{cases} 1, & \hat{P} \geq 0.5 \\ 0, & \hat{P} < 0.5 \end{cases}$$

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